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Risk assessment in social lending via random forests

Milad Malekipirbazari, Vural Aksakalli *

Istanbul Sehir University, Istanbul 34662, Turkey

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ABSTRACT

With the advance of electronic commerce and social platforms, social lending (also known as peer-to-peer lending) has emerged as a viable platform where lenders and borrowers can do business without the help of institutional intermediaries such as banks. Social lending has gained significant momentum recently, with some platforms reaching multi-billion dollar loan circulation in a short amount of time. On the other hand, sustainability and possible widespread adoption of such platforms depend heavily on reliable risk attribution to individual borrowers. For this purpose, we propose a random forest (RF) based classification method for predicting borrower status. Our results on data from the popular social lending platform Lending Club (LC) indicate the RF-based method outperforms the FICO credit scores as well as LC grades in identification of good borrowers.

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1. Introduction

Social lending, also known as peer-to-peer (P2P) lending, is emerging as an alternative to banks where individual members lend and borrow money using an online trading platform without the help of official financial intermediaries such as banks. The attractive feature of doing business on a peered platform is the higher potential of mutual profitability. Borrowers can obtain loans at lower interest rates and lenders can loan money at better rates than what they can get from a bank. In particular, via social lending, lenders can find a multitude of potential borrowers and choose among them the ones they wish to lend. Since the ultimate savers are predominantly consumers, and consumers are the individuals who are actually lending in the social lending model, there is no need to increase the liquidity of the loans by securitizing them. Since social lending is powered by the Internet, it would not take much effort to connect small communities such as towns, religious, or ethnic groups for the purpose of intra-community lending and borrowing.

The popular social lending platforms currently in use today are the U.S.-based Prosper[1] and Lending Club Corp.,[2] UK-based Zopa Ltd.,[3] and Germany-based Smava GmbH.[4] All of these social platforms rely on the credit scores provided by a cooperating credit reporting agency; Experian, TransUnion LLC, Equifax Inc., and Schufa Holding AG respectively. The popularity of these platforms is growing as recently indicated by Lending Club (LC) which has reached 6.2 billion USD in total loans by January 2015 and transformed into a 8.5 billion USD publicly-traded company, becoming the world's largest social lending platform (LendingClub.com, 2015).

Our standpoint in this work, which is consistent with other studies in this context, is that even though social lenders can base their investment strategy on the traditional financial credit scores provided by external agencies, available data suggest that social lending tends to have different dynamics when compared to traditional lending. For instance, the distribution of lenders' bids on social loan listings when indexed by time follows a power law (Rodgers & Zheng, 2002), an indication of a herding behavior. Assume that lenders and loan listings are denoted by nodes and an edge between them denotes that the lender is interested in the corresponding listing. Since the distribution of bids indicates a bias towards highly connected nodes, this in reality means that once a loan listing has a hundred or more lenders bid on it, then that specific listing is more likely to attract more and more lenders. This in turn makes the corresponding listing more likely to get funded in the end due to high lender interest.

A comprehensive analysis of LC loan data by Emekter, Tu, Jirasakuldech, and Lud (2015) reveals two key findings:

1. There exists a selection bias in the sense that high-income borrowers with the highest FICO credit scores[5] do not borrow from LC. In particular, top one third of the consumers with respect to FICO scores do not create any loan listings on LC.

* Corresponding author.
E-mail addresses: miladmalekipirbazari@std.sehir.edu.tr (M. Malekipirbazari), aksakalli@sehir.edu.tr (V. Aksakalli).

[1] http://www.prosper.com.
[2] http://www.lendingclub.com.
[3] http://www.zopa.com.
[4] http://www.smava.de.

[5] FICO, a publicly traded corporation, produces scoring models that are most commonly used and distributed by TransUnion, Equifax, and Experian.

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2. Higher interest rates charged on the higher risk borrowers are not worth the risk. Specifically, higher rates charged for the borrowers with low LC's own credit grade are not high enough to overcome the greater default risk that the lenders take.

The above two findings imply that, from a profitability point of view, identifying the "good borrowers", i.e., those who will pay back their loan in full within due time, is of great importance for investors participating in social lending. Profitability of social investors, on the other hand, is a critical component in continued interest in social lending as well as overall sustainability of the social lending market. In this regard, subsequent to a risk and return efficiency analysis, Emekter et al. (2015) suggest that "the lenders would be better off to lend only to the safest borrowers with the highest LC grades". Despite this suggestion, we show in this work that even borrowers with the highest FICO scores or LC grades are not necessarily good borrowers, which in turn indicate that traditional financial score metrics are not well-equipped to capture the non-conventional dynamics prevalent in social lending.

In order to improve identification of good borrowers within the context of social lending, this study proposes and presents comparisons of different machine learning methods including random forests (RFs), support vector machines (SVM), logistic regression (LR), and *k*-nearest neighbor (*k*-NN) classifiers. Our computational results on LC data between January 2012 and September 2014 for a total loan amount of about one billion USD indicate that random forests outperform the other classification methods and stand as a scalable and powerful approach for predicting borrower status. In fact, an empirical comparison reveals that RFs significantly outperform both FICO scores and LC grades in identification of the best borrowers in terms of low default probability.

The rest of this manuscript is organized as follows: Section 2 provides a brief literature review on social lending. In Section 3, we introduce basics of social lending, describe the financial features used in prediction, and provide an exploratory data analysis. Section 4 presents the classifiers and Section 5 provides a comprehensive experimental comparison. Our summary and conclusions are presented in Section 6.

2. Related work

An application of machine learning principles in social lending is the use of Gaussian mixture models on the Prosper data set containing loan transactions between November 2005 and December 2008 (Lopez, 2009). An interesting finding therein is that if an individual with a high-risk FICO score belongs to a trusted social community, then this individual's social membership can still help secure a loan. Thus, even though a high-risk credit score usually means lack of access to traditional bank-mediated financial markets, a positive social feature can outweigh a highly negative financial feature in socially mediated markets.

Complex behavioral dynamics further complicate the social lending process. For example, the simple auction mechanism used in some social lending platforms can lead to unpredictable payments for the borrower. An incentive compatible mechanism might be more suitable to eliminate this inefficiency where lenders report their true interest rate and do not change their rate dynamically (Chen, Ghosh, & Lambert, 2009). Otherwise, such inefficiencies enable users with adversarial interests to use the lending platform as an arbitrage opportunity: borrow at 10% and then loan at 20% (Steelman, 2006).

The notion of groups was introduced into social lending with Prosper. Users of this platform can form groups around an affinity that all members share such as a certain topical interest, geographic location, a peer group, or simply around the reasons to borrow. Groups have leaders that act as mediators of loan activity. This mediation can be in the form of pre-evaluating group members, endorsing potential borrowers, inverting and diffusing the risk of a particular group member default among all group members, or encouraging all members to proactively screen new members and apply peer pressure. Empirical studies show that when a group leader in a lending platform mediates the group actively, the risk factor drops considerably. In addition, if a group leader recommends a loan listing put together by one of the group members, this endorsement increases the chance of the loan being issued and also decreases the final interest rate (Berger & Gleisner, 2009).

There exist several studies proposing a set of guidelines in order to make purely rational investment decisions in social lending. In one such study on Prosper loan data that includes loan transactions between November 2005 and March 2007, irrespective of the financial credit rating categories,[6] three simple rules help decrease the risk of a default (Klafft, 2008). These investment rules are as follows:

1. Invest only in borrowers without any delinquent accounts.
2. Invest only in borrowers that satisfy Rule 1 and that have a debt-to-income (DTI) ratio less than 20%.
3. Invest in borrowers that satisfy Rule 2 and that have no credit inquiry reports during the last 6 months.

In studies conducted on social communities, herding (denser clustering following a power law regime) effects usually prevail (Gao & Feng, 2014; Lee & Lee, 2012; Yum, Lee, & Chae, 2012). Empirical studies show that the tendency of an individual to join a given community is effected by the number of friends in this community and the inter-connectedness of this individual's friends within the community. Such behavioral bias also exist in investment decisions of lenders at Prosper. The loan data between 2006 and 2008 show that previous lender decisions effected subsequent lender decisions and lender decisions were not made purely rationally (Shen, Krumme, & Lippman, 2010). For the interested reader, there exist other real-world networks (such as airports and power grid transmission lines) and other social networks (such as DBLP and LiveJournal) that also exhibit a herding behavior (Amaral, Scala, Barthelemy, & Stanley, 2000; Backstrom, Huttenlocher, Kleinberg, & Lan, 2006).

The closest study to ours is the work of Emekter et al. (2015) where the authors analyze LC data between May 2007 and June 2012 and present a logistic regression (LR) model for predicting default probability of a borrower. Their model includes FICO scores as well as LC grades in default prediction. In contrast, our study uses all the available financial features other than the FICO scores and LC grades in order to assess the relevance and prediction power of these two metrics in social lending. Nonetheless, we show in Section 4 that one can get much better prediction accuracy using random forests compared to LR even with the exact same features used in building this LR model.

3. Overview and data analysis

3.1. Social lending overview

The LC social lending platform works as follows:

[6] Prosper grades its individual platform users into credit grade buckets in the increasing risk order as AA, A, B, C, D, E, and HR (high-risk) depending purely on credit scores assigned by Experian.

- Eligible borrowers with respect to LC's selection criteria create a loan listing.
- LC determines an interest rate for the loan based on the borrower's LC credit grade.
- Lenders have access to borrowers' financial information such as FICO score, LC grade, debt to income ratio, home ownership status, and number of delinquent accounts for their evaluation of potential borrowers. A loan listing includes other details such as the reason to borrow and relevant demographic information for lenders to review.
- Lenders assume the risk of defaults. Thus, a common lending strategy is to spread out a certain amount of money across a large number of borrowers, thereby reducing the probability of a loss.
- For a loan to be granted, the borrower needs to find enough lenders to cover the entire loan amount.
- A loan listing either expires without enough interest or it gets fully funded before the expiration date.
- LC collects a certain service fee from lenders for any payments they receive from the borrowers.

3.2. Explanatory data analysis

The data set used in our study was retrieved from LC website LendingClub.com for the period between January 2012 and September 2014. The data contain approximately 350 K borrower records with all the credit information normally used by credit agencies to assign FICO scores. Since many of these records are loans that have not been issued or reached maturity yet, and thus do not contain information about the empirical creditworthiness of the borrowers, we filtered the data to only include loans with a status of fully-paid or defaulted. There were about 68 K such loans, which correspond to a total loan amount of about one billion USD.

A critical first step in any machine learning application is selection of features[7] with good predictive power of the response variable. LC data contain a total of 35 financial and other borrower-specific features for each loan record. Upon a careful analysis, we selected 23 of these features to be used in predictive modeling. Next, we generated a total of 15 features to be used in our models subsequent to certain data manipulations such as taking ratios of the original attributes, using logarithms to compress exponentially-distributed data, converting dates to time lengths, and eliminating outliers. Data pre-processing and manipulation tasks were conducted using the open-source statistical software R whereas the machine learning tasks were carried out with the open-source machine learning software WEKA. Features used in our predictive models are described below.

- **Loan Status (Response Variable)**: Binary variable indicating whether the borrower defaulted on the loan or fully paid off the loan. For convenience, the defaulted loans shall be referred to as "bad" and fully paid ones as "good" loans.
- **Annual Income**: The annual income information provided by the borrower during registration. Data manipulation: the natural logarithm function.
- **Credit Age**: Date of the earliest credit line opened by borrower, converted to months. Data manipulation: the natural logarithm function.
- **Delinquencies**: The number of delinquencies in the last two years for the borrower. Data manipulation: right-censor $\geqslant 2$. That is, values greater than 2 were set to 2.
- **Employment Length**: Employment length in years. Possible values are integers between 0 and 10 with 0 meaning less than one year and 10 meaning ten or more years.
- **Home Ownership**: The home ownership status information provided by the borrower during registration. The possible values are: Rent, Own, and Mortgage.
- **Inquiries**: The number of credit inquiries in the last six months on the borrower. Data manipulation: right-censor $\geqslant 3$.
- **Loan Amount**: Dollar amount of the loan. This amount cannot exceed $35,000.
- **Loan Purpose**: A category provided by the borrower for the loan request. Possible values are: Debt Consolidation, Home Improvement, Credit Card, Moving, Small Business, Car, Major Purchase, Vacation, Medical, Renewable Energy, House, Wedding, and Other.
- **Open Accounts**: The number of open credit lines on the borrower's credit file.
- **Total Accounts**: The total number of credit lines currently on the borrower's credit file.
- **Term**: The number of monthly payments on the loan. Values can be either 36 months or 60 months.

Pie charts of some of the features above are shown in Fig. 1 whereas histograms and box plots of the other attributes are displayed in Fig. 2. Next, we describe the features that are simple ratios of other features. These ratios represent a meta-normalization of certain borrower characteristics that otherwise would not be easily captured. These ratio features are explained below and displayed in Fig. 3.

- **DTI (Debt to Income Ratio)**: Ratio of the borrower's total monthly debt payments to the borrower's monthly income.
- **Income to Payment Ratio**: Ratio of the loan's monthly payments to monthly income. Data manipulation: the natural logarithm function. This is a non-standard financial feature that we introduce in this study. The idea behind this feature is that a monthly payment of $500 may be inconsequential for someone who makes $10,000 per month, but it would be a life-changer for someone who makes $1000 per month. Considered separately, it might be difficult for a machine learning algorithm to capture the essence of this attribute.
- **Revolving Utilization Rate**: The amount of credit the borrower is using relative to all available revolving credit (i.e., credits that do not have a fixed number of payments such as credit cards).
- **Revolving to Income Ratio**: Ratio of revolving credit balance to the borrower's monthly income. Data manipulation: the natural logarithm function. This is another non-standard financial feature that we introduce in this study.

Two additional attributes that are present in the LC data are FICO scores and LC grades, which are described below. Our primary purpose in this study is to assess relevance and prediction power of these two attributes regarding borrower risk attribution and therefore they are excluded from our predictive models.

- **FICO Score**: FICO scores are the standard credit scores that are used in majority of the lending decisions in the US. They are calculated from various financial attributes derived from the borrowers' credit records. In the LC data, FICO scores are reported as two numbers in the form of FICO low and FICO high. We average these two numbers and call it the FICO score. Histograms and box plots of the FICO scores are shown in Fig. 4a.
- **LC Grade**: LC assigns a grade from A to G to each loan using a proprietary algorithm based on the loan characteristics and risk assessment of the borrower. These seven grades are further divided into 5 levels resulting in loan grades of A1 through G5

[7] In this manuscript, the terms feature, attribute, and predictor are used synonymously.

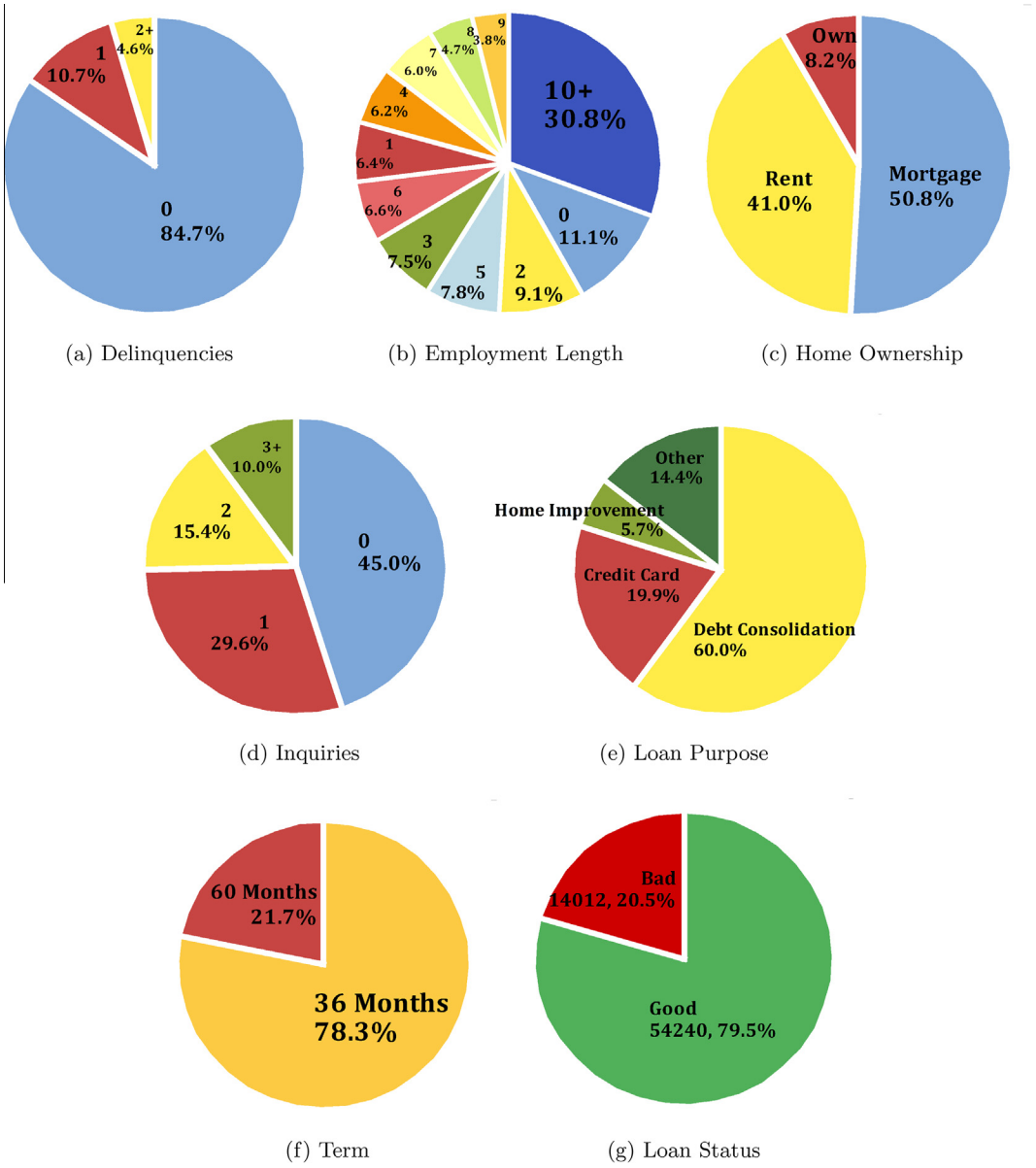


Fig. 1. Pie charts of delinquencies, employment length, home ownership, inquiries, loan purpose, term and loan status.

with A1 corresponding to the safest loan and G5 corresponding to the most risky loan. LC assigns interest rates to each loan ranging from 7% to 25% such that the interest rate monotonically increases as the loan grade decreases. For convenience, we converted these grades to numbers between 1 and 35 where 35 corresponds to A1 (highest grade) and 1 corresponds to G5 (lowest grade). Histograms and box plots of the LC grades are depicted in Fig. 4b.

3.3. Statistical significance of features

Out of the 15 features we consider, 12 of them are numeric. These attributes were standardized to have zero mean and unit standard deviation before any model building process. The remaining 3 attributes are nominal: Home Ownership (3 levels), Loan Purpose (13 levels), and Term (2 levels). Prior to building the first three machine learning models; namely, *k*-NN, LR, and SVM; the nominal attributes were binarized, in which case we were left with a total of $12+3+13+2=30$ numeric attributes. On the other hand, in the case of RFs, the nominal attributes were untouched.

For a better understanding of the data and the attributes, we computed Information Gain and correlation for numeric and binarized nominal attributes with respect to the loan status. These two statistics are displayed in Table 1 for the top 15 attributes. The two statistics result in similar rankings with LC grade being the highest followed by Income to Payment Ratio, Annual Income, FICO score, and DTI. Nonetheless, even the LC grade, which solely determines the interest rate for a particular loan, has a correlation less than 0.02 and an Information Gain less than 0.03. We observe in general that, when considered separately, the available attributes are quite weakly correlated with the loan status.

4. Methodology

4.1. The mathematical model

In line with the notation in James, Witten, Hastie, and Tibshirani (2013), we model the borrower status prediction problem as follows: Let the binary response variable be denoted by Y such that

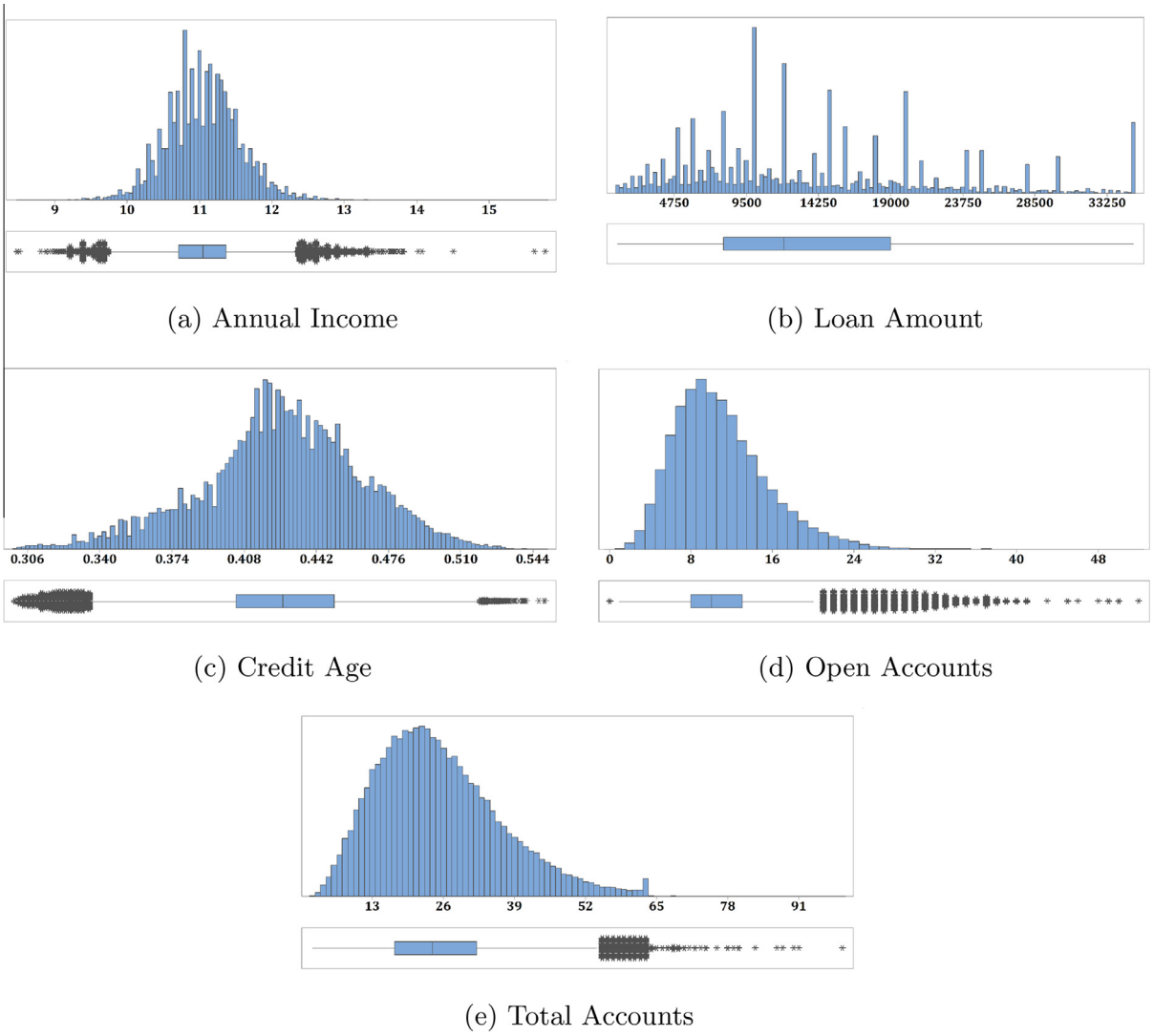


Fig. 2. Histograms and box plots of (log) annual income, loan amount, (log) credit age, open accounts, and total accounts.

$$Y = \begin{cases} -1, & \text{if Bad Borrower} \\ +1, & \text{if Good Borrower.} \end{cases}$$

Our mathematical model is given as

$$Y = f(\mathbf{X}) + \epsilon$$

where $\mathbf{X} = (X_1, \ldots, X_p)$ is the feature vector, p is the number of features, and ϵ is the irreducible error term in the model that captures measurement errors and other noise in the data. Note that even though the function f is assumed to exist, it is almost never known in practice. In addition, even if f is known, there would still be prediction errors due to the fact at each $\mathbf{X} = \mathbf{x}$, there is a distribution of possible Y values in general. Thus, the function f is defined as

$$f(\mathbf{x}) := E(Y|\mathbf{X} = \mathbf{x})$$

where the right-hand-side (RHS) is the expected value of Y for a particular realization $\mathbf{x}$ of the feature vector $\mathbf{X}$. Rather than assuming a binary value, $f(\mathbf{x})$ specifies the probability of a borrower being a good borrower, which can be expressed as

$$f(\mathbf{x}) = E(Y|\mathbf{X} = \mathbf{x}) = Pr(Y = 1|\mathbf{X} = \mathbf{x}) \quad (1)$$

An important property of $f(\mathbf{x})$ is that it is the function that minimizes $E[(Y - g(\mathbf{X}))^2|\mathbf{X} = \mathbf{x}]$ over all functions g at all points $\mathbf{X} = \mathbf{x}$. That is,

$$f(x) = \arg \min_g E[(Y - g(\mathbf{X}))^2|\mathbf{X} = \mathbf{x}]$$

Our goal in this study is to find a good estimate of $f(\mathbf{x})$, which we denote by $\hat{f}(\mathbf{x})$. For any $\hat{f}(\mathbf{x})$, it holds that

$$E[(Y - \hat{f}(\mathbf{X}))^2|\mathbf{X} = \mathbf{x}] = [f(\mathbf{x}) - \hat{f}(\mathbf{x})]^2 + Var(\epsilon)$$

where the first RHS term denotes the reducible error and the second RHS term denotes the irreducible error that is inherent in Y. Thus, the learning problem at hand is to find a good estimate $\hat{f}^*(\mathbf{x})$ that minimizes the reducible error, i.e., $[f(\mathbf{x} - \hat{f}(\mathbf{x})]^2$. We consider a total of four different classifiers for this task, which are described below.

4.2. k-Nearest neighbors (k-NN)

The algorithm of k-nearest neighbors (k-NN) is a very simple, yet popular and powerful non-parametric method widely used in classification. The inputs to the algorithm are the k nearest training instances to the test instance with respect to a certain distance function, typically with a small k between 1 and 10. The classification is based on a majority vote of these k nearest neighbors. It might sometimes be helpful to weight the neighbors' contributions such that the closer neighbors have more effect on the final deci-



Fig. 3. Histograms and box plots of DTI, Income to payment ratio, revolving utilization rate, and revolving to income ratio.

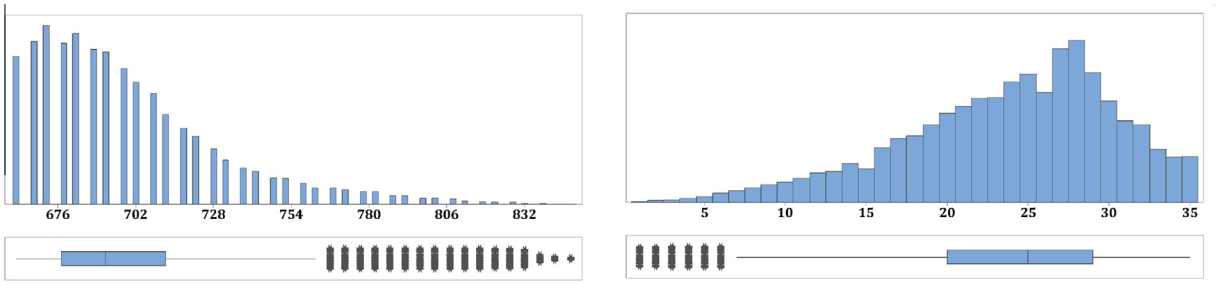


Fig. 4. Histograms and box plots of FICO scores and LC grades.

Table 1
Information Gain and correlation with respect to loan status.

Rank	Attribute	Information gain	Correlation
1	LC Grade	0.0294	0.198
2	Income to payment ratio	0.0191	0.143
3	Annual income	0.0126	0.126
4	FICO Score	0.0124	0.125
5	DTI	0.0102	−0.119
6	Revolving line utilized	0.0092	−0.111
7	Term	0.0074	−0.104
8	Revolving to income ratio	0.0071	0.079
9	Total accounts	0.0042	0.073
10	Home ownership: mortgage	0.0041	0.075
11	Home ownership: rent	0.0037	−0.072
12	Loan amount	0.0026	−0.044
13	Loan purpose: small business	0.0019	−0.054
14	Inquiries	0.0018	0.002
15	Credit age	0.0017	0.044

sion that the farther ones (Aha & Kibler, 1991). In our setting, the estimate $\hat{f}_{k-\mathrm{NN}}(\mathbf{x})$ of k-NN can be expressed as

$$\hat{f}_{k-\mathrm{NN}}(\mathbf{x}) := \mathrm{Majority}(Y|\mathbf{X} \in \mathscr{N}_k(\mathbf{x}))$$

where *Majority* denotes the majority vote function and $\mathscr{N}_k$ denotes the k closest neighbors of $\mathbf{x}$ with respect to the Euclidean distance in the p-dimensional space. In our implementation, we take $k = 1$ for simplicity and convenience.

4.3. Logistic regression (LR)

A (simple) linear regression estimator $\hat{f}_{\mathrm{L}}$ is defined as a linear combination of the individual attributes in the form

$$\hat{f}_{\mathrm{L}}(\mathbf{x}) := \sum_{i=0}^{p} \beta_i x_i = \boldsymbol{\beta}' \mathbf{x} (x_0 = 1)$$

where the estimates $\hat{\beta}_i$ of β_i are computed via a least-squares fit for $i = 0, \ldots, p$ using the observations in the training data. A more appropriate technique for binary classification is the logistic regression estimator below:

$$\hat{f}_{\mathrm{LR}}(\mathbf{x}) := \frac{e^{\beta' \mathbf{x}}}{1 + e^{\beta' \mathbf{x}}}$$

In LR, the estimates $\hat{\beta}_i$ are calculated using the maximum likelihood method (Landwehr, Hall, & Frank, 2005; Sumner, Frank, & Hall, 2005). A useful property of LR is that it guarantees an output value between 0 and 1 that can be interpreted as class-conditional probabilities in classification problems.

4.4. Support vector machines (SVM)

For a data set with a binary response variable, support vector machines separate the data into two regions (one for each class) in the p-dimensional feature space via a hyperplane with the maximum margin width between instances of the two classes (Cortes & Vapnik, 1995). This process is illustrated in Fig. 5 for two different cases in two-dimensional space. Maximizing the margin width decreases the complexity of the model and the overall risk of errors. When the data is not separable by a hyperplane, which is usually the case in practice, a soft margin is used. In this situation, a positive slack is added to the instances on the wrong side of the margin. This slack increases proportional to how far the corresponding instance is from the margin. The goal is to minimize the sum of these slacks while maximizing the width of the margin. A regularization parameter C governs the relative cost of each objective in the optimization process, which is taken as 1.0 in our implementation. The SVM optimization problem can be stated as a quadratic programming problem in the dual form as

$$\max_{\alpha_l} \left(\sum_{l=1}^{n} \alpha_l + \frac{1}{2} \sum_{l,r=1}^{n} y_l y_r \alpha_l \alpha_r K(x_l, x_r) \right)$$

subject to the constraints $0 \leqslant \alpha_l \leqslant C$ for training instances $l = 1$–n and, $\sum_{l=1}^{n} y_l \alpha_l = 0$ where α_l is the Lagrange multiplier associated with the instance l. In this study, we employ the quadratic polynomial kernel $K(x_l, x_r) = (1 + x_l . x_r)^2$. We chose SVMs as a candidate classifier as they haven been used successfully in similar applications before (Han, Han, & Zhao, 2013; Schebesch & Stecking, 2005a; Zhou, Lai, & Yu, 2010).

4.5. Random forests (RF)

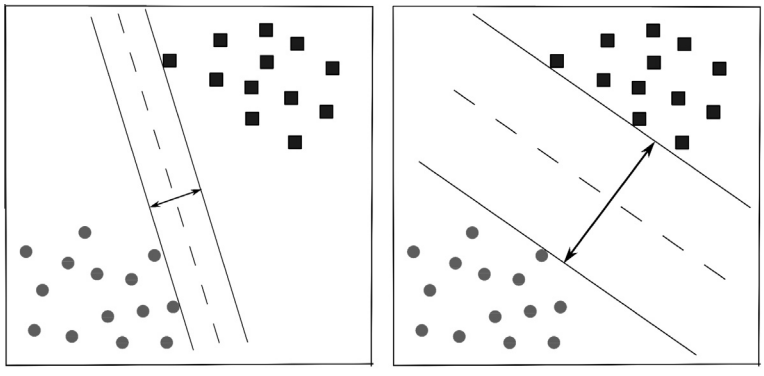
Decision tree learning is a popular classification method that grows a tree-based structure with class-conditional probabilities at the end of the tree branches. The decision tree starts with a root node and gradually builds sub-trees with internal nodes that are connected by emanating branches and ends with terminal nodes called leaves. Each internal node corresponds to a test of a feature (e.g., a borrower owns a house or otherwise) and branches represent a binary partition of the test attribute. The process of building a decision tree is a divide-and-conquer approach in the sense that the root node corresponds to the entire training data and each node split corresponds to a partitioning of the available data at that node based on the test condition for the associated feature. There are two critical issues in decision tree learning: (1) how to choose the split attribute at each internal node and (2) how many levels to have at each tree branch, i.e., when to stop splitting. Within the context of random forests, which are collections of decision trees, splitting is done with respect to *Gini Index*, which is described below, and number of levels in each tree branch is controlled by an algorithm parameter d (Breiman, 2001).

The Gini Index at an internal tree node is calculated as follows: For a candidate (nominal) split attribute X_i, denote possible levels as $L_1, \ldots, L_J$. Gini Index for this attribute is calculated as

$$G(X_i) := \sum_{j=1}^{J} Pr(X_i = L_j)(1 - Pr(X_i = L_j)) = 1 - \sum_{j=1}^{J} Pr(X_i = L_j)^2.$$

Once Gini Indices are computed for each candidate split attribute, the split is done on the attribute that has the highest Gini Index.

Decision tree classifiers have several attractive properties: they are easy to interpret, they can handle both numerical and nominal data, and they are easy to build. Nonetheless, decision trees are not always competitive with other classification techniques. Thus, in order to improve the accuracy of trees, one sometimes need to employ ensemble methods such as boosting (iterative learning from misclassified instances) and bagging (building multiple trees and combining the results). Random forests (RFs), which can be seen as an enhanced bagging technique, is a powerful method for constructing a forest of random decision trees. RFs de-correlate the decision trees in the forest via randomization of split attributes that leads to an improvement over traditionally bagged trees and reduces the variance when averaged over the trees (Breiman, 2001). RFs also build multiple decision trees on bootstrapped training samples. However, while building these trees, the candidate split attributes in each tree are chosen by a random selection of m attributes from the full set of p attributes, as illustrated in Fig. 6. The split is allowed to use only one of these m attributes and a fresh selection of m attributes is made at each split. In each tree, splitting is continued until the tree reaches a depth of d. In our implementation of RFs, we chose a forest size of 80 with $m = 5$ and $d = 25$. Fine-tuning of these parameters is described in Section 4.8.



(a) Small margin (b) Large margin

Fig. 5. Separating hyperplanes with small and large margins in SVM.

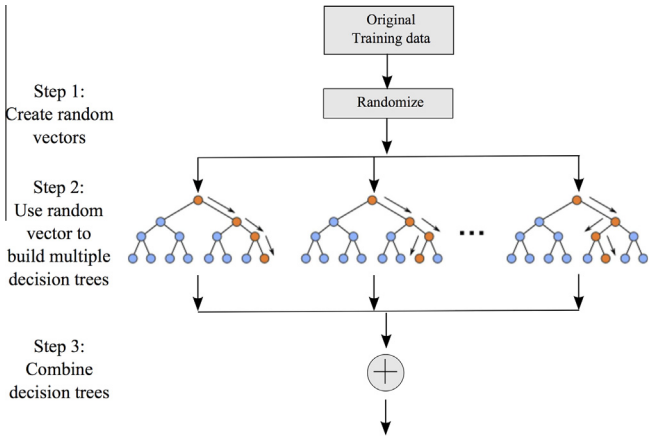


Fig. 6. Demonstration of the random forest methodology.

4.6. Cost sensitive analysis

Since accepting a bad borrower carries much greater risk than rejecting a good borrower, we use a weighted cost matrix in our models to increase the cost of misclassification associated with bad borrowers. As suggested in Schebesch and Stecking (2005b), we performed all our experiments in this study with a 5-to-1 cost ratio such that misclassification of a bad borrower (as a good borrower) is 5 times more costly than misclassification of a good borrower (as a bad borrower). For this purpose, we used the *CostSensitiveClassifier* meta-function within WEKA.

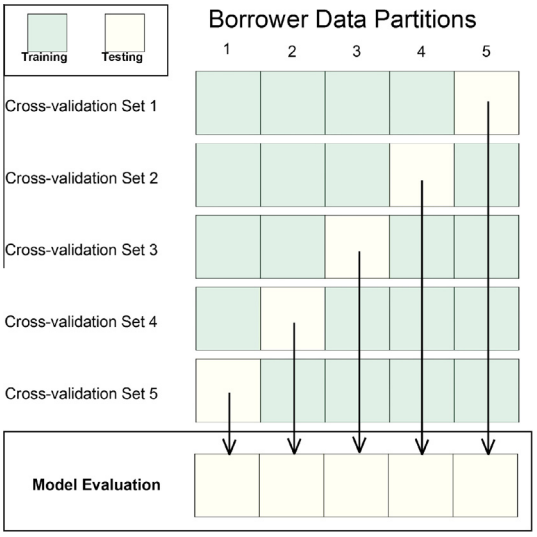


Fig. 7. Demonstration of 5-fold cross validation.

4.7. Model evaluation and assessment

For evaluation of the candidate classifiers, we employed the popular 5-fold cross-validation (CV) that has been shown to provide a good trade-off between model over-fitting and under-fitting in general (Huang, Chen, & Wang, 2007). Thus, the data set of 68 K loans was first partitioned into five equally-sized slices. For each one of the 5 CV folds, one slice was set aside for testing and the remaining 4 slices were used for training the classifiers. This process is illustrated in Fig. 7. CV specifically enables us to test the performance of a model on every instance in the available data set without having used it in the training phase.

Regarding performance assessment of the classifiers, we took an average over the 5 folds in CV of the following 7 performance metrics: (1) overall classification accuracy rate on the test slice in the fold, (2) the usual area under the Receiving Operating Characteristic (ROC) curve (AUC) for the good class in the test slice, (3) root mean square error (RMSE), (4)–(7) the confusion matrix: true positive (TP) and false positive (FP) rates for the good and bad classes respectively in the test slice.

4.8. RF parameter fine-tuning

This section investigates the effects of forest size, the number of split features m, and the maximum tree depth d on the performance of RFs in our problem so that we can identify the optimal values for these parameters. For simplicity and convenience, this fine-tuning process was conducted in two steps: (1) determining the optimal forest size and, (2) determining the optimal m and d values. First, by using the "auto" option in WEKA for m and d, we constructed forests with sizes ranging from 1 to 500 trees in increments of 10 trees. The comparison results are shown in Fig. 8 where we observe that building trees beyond 80 did not result in considerable additional performance, yet increased the run time considerably. Thus, we settled on a forest size of 80 as a reasonable trade-off between execution time and classification performance.

The suggested value for the number of split features m is $\lceil \log_2(p) \rceil$ (Breiman, 2001), which in our case corresponds to 5. In order to identify a good value for m, we considered $m = 5,\ 10,\ 20$ and for each m value, we experimented with d values ranging from 1 to 40. For each $m,\ d$ combination considered, we used a forest size of 80. Experimental results shown in Fig. 9

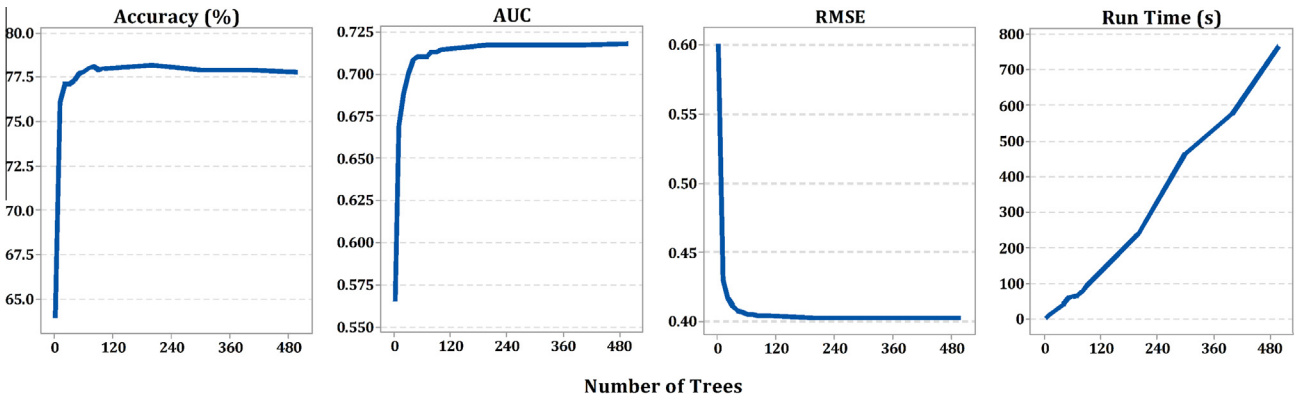
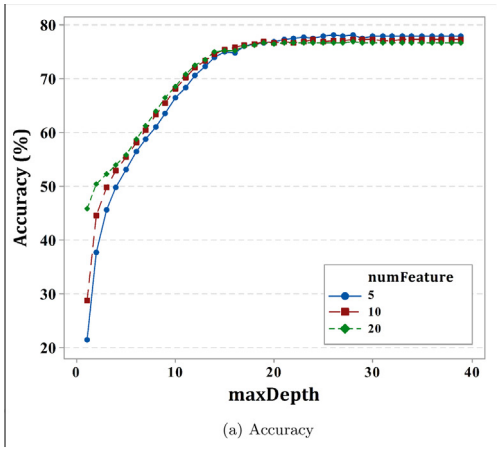
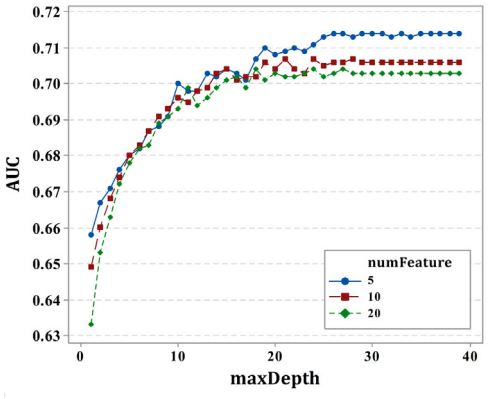


Fig. 8. RF performance with respect to the forest size. Run times shown correspond to the total time for the 5-fold CV process for each size.



(a) Accuracy



(b) AUC

Fig. 9. RF performance with respect to number of split features and maximum depth.

Table 2
Performance comparison of the classifiers. AUC denotes area under the ROC curve for the good class, RMSE the root mean square error, TP true positive, and FP false positive respectively.

Rank	Classifier	Accuracy (%)	AUC	RMSE	TP Rate		FP Rate	
					Good	Bad	Good	Bad
1	Random forest	78.0	0.71	0.42	0.88	0.31	0.69	0.13
2	Nearest neighbor	70.1	0.53	0.55	0.82	0.25	0.74	0.18
3	Support vector machine	63.3	0.62	0.68	0.47	0.78	0.22	0.53
4	Logistic regression	54.5	0.68	0.51	0.49	0.77	0.23	0.51

indicate that (1) 5 split features does indeed exhibit the best performance and (2) accuracy rates and AUCs practically flatten out after a tree depth of 25 regardless of the m value. Thus, in our implementation, we settled on a tree size of 80 with $m = 5$ and $d = 25$.

5. Experimental results

5.1. Comparison of the classifiers

This section presents comparisons of the classifiers on the LC data as evaluated via 5-fold CV. The results are shown in Table 2 and Fig. 10 respectively. Overall, we observe that RFs have the highest accuracy rate at 78.0% and the highest AUC at 0.71 with the lowest RMSE at 0.42. Whereas k-NN has the second highest accuracy rate at 70.1%, LR has the second highest AUC at 0.68. Based on these results, we declare RFs to be the best overall classifier compared to the other three alternatives.

5.2. Comparison against FICO scores and LC grades

We now compare the performance of RFs against FICO scores and LC grades. Our methodology in this comparison is to compute thresholds for all three metrics that result in identical acceptance rates and then compare the ratio of the number of defaults to the number of loans corresponding to that acceptance rate. For instance, if we were to accept only borrowers with a FICO score above 750, we would be accepting about 8% of borrowers. This specific subset of accepted borrowers has a default rate of 8.2%. Then, we can perform a threshold sweep over the RF scores to find the threshold that corresponds to exactly 8% acceptance rate. The RF classifier has only 3.1% of the borrowers defaulting in this subset. In this way, we can declare RF scores to be superior to FICO scores at predicting creditworthiness in the top 8% acceptance rate.

Comparison results for acceptance rates ranging from 0.1% to 20% for all three metrics are shown in Fig. 11. Within the 3% acceptance rate, we see that RFs do not misclassify any bad borrowers as good borrowers. In particular, we observe that up to the 10% acceptance rate, RFs outperform the other two metrics in identification of good borrowers. However, this superior performance of RFs for the top 10% of the borrowers comes at the cost of misclassifying good borrowers (as bad borrowers) outside of this acceptance rate. Specifically, beyond the 10% acceptance mark, even though RFs are still superior to FICO scores, LC grades show better performance.

5.3. Comparison against existing methods

This section compares our RF methodology against the current state-of-the-art in LC loan status prediction in the literature, which is the logistic regression model in Emekter et al. (2015). This model advocates utilization of only four attributes: FICO score, LC score, DTI, and Revolving Utilization Rate. As before, we trained the 4-feature LR with a 5-to-1 cost ratio for fairness. Table 3 shows that all-attribute RF (except FICO scores and LC grades) significantly outperforms the 4-feature LR of Emekter et al. (2015) with an accuracy of 78.0% vs. 51.0%. In particular, the 4-feature LR's misclassification of bad borrowers (as good) is 56% whereas this ratio is only 13% for all-feature RF.

We also considered the question if one would be better served by using RFs with the above 4 features as opposed to LR. This com-

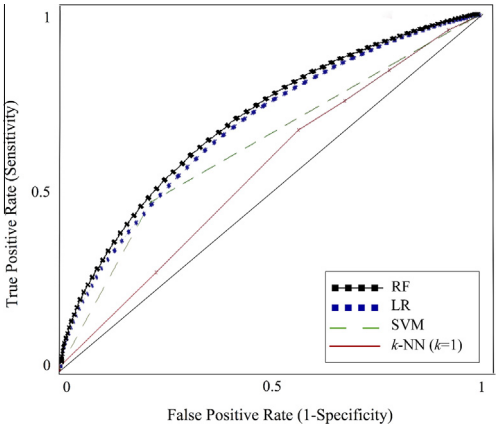


Fig. 10. Comparison of ROC curves for different classifiers.

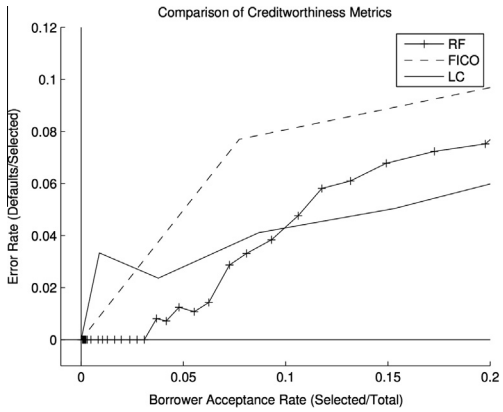


Fig. 11. Comparison of creditworthiness metrics in terms of error rate vs. borrower acceptance rate.

Table 3
Comparison of RF and LR with the features FICO score, LC grade, DTI, and Revolving Utilization Rate against RF with all the feature (except FICO scores and LC grades). The first row is taken directly from Table 2.

Classifier	Accuracy (%)	AUC	RMSE	TP Rate		FP Rate	
				Good	Bad	Good	Bad
(All-feature) random forest	78.0	0.71	0.42	0.88	0.31	0.69	0.13
(4-Feature) random forest	69.8	0.61	0.45	0.79	0.32	0.68	0.20
(4-Feature) logistic regression	51.0	0.65	0.52	0.45	0.76	0.24	0.56

parison is also given in Table 3. It can be seen that 4-feature RF achieves an overall accuracy of 69.8% compared to 51.0% accuracy of 4-feature LR. In addition, the 4-feature RF's misclassification rate for bad borrowers is 20% compared to 56% rate of the 4-feature LR. Thus, we conclude that even with the same 4 features considered in Emekter et al. (2015), RFs exhibit superior performance compared to LR.

6. Summary and conclusions

Identifying true creditworthiness of a potential loan borrower is vital for healthy functioning of social lending markets. In order to compute the risk score of an individual, financial features such as past financial history, existence of delinquent accounts, debt to income ratio (DTI) and various other financial features are used. This study presents an RF-based methodology for identification of good borrowers in social lending using the world's largest social lending platform LendingClub.com's publicly available historical records. We introduce non-standard financial features in order to increase reliability of the computed risk scores, and we propose and present a comparison of the machine learning methods RF, SVM, LR, and *k*-NN for identifying good borrowers in social lending. Our computational results indicate that RFs outperform the other classifiers as well as the FICO scores and LC grades in predicting good customers.

A limitation of our approach is that while RFs are quite powerful in predicting good borrower status, this comes at the cost of misclassifying some of the good borrowers as bad who are not the best (i.e., those not on top when ranked based on RF scores). In particular, RFs are superior in identifying best of the best borrowers, but they exhibit a decrease in relative performance beyond the 10% acceptance rate mark. Nonetheless, as mentioned earlier, it has been shown that high risk borrowers are not worth the higher returns in general from a risk/expected return trade-off point of view. Consequently, RFs stand as a more logical choice for potential LC lenders for making investment decisions. In particular, a lender can simply choose a borrower, say, in the top 3% of the population with respect to RF scores, and be confident that even though the returns might not be as high, there is practically no risk of a default.

In this study, we do not consider any adjustments to interest rates for a trade-off analysis between default risk and higher returns. Hypothetically, should LC allow for the loan interest rates to be determined in a free market via an auctioning mechanism, lenders would likely be less hard-pressed to loan only to best of the best borrowers as the much higher rates might outweigh relatively high default risks. We believe there is a considerable amount of room in analysis of dynamic interest rates in the LC platform and optimal balancing of risk and return trade-offs based on lenders' risk preferences, which constitute a rather important direction for future research.

Another research direction is integration of social media data for better borrower risk assessment in social lending. There exist a vast amount of data on borrower characteristics on social networks such as Facebook and Twitter. We believe that there is a significant potential in mining of social media data for more accurate borrower risk attribution and thereby integration of social media into social lending. One other direction for future research is further fine-tuning of the SVM parameters (including trying different kernel functions) for loan status prediction. This fine-tuning process is a rather time-consuming task as it requires a careful search in multi-dimensional parameter space. In our implementation, we conducted only a limited amount of experiments for SVM fine-tuning. It is plausible that a carefully tuned SVM can show similar performance to RFs for classification.

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